

Paper Replication Report #014: Type I Disk Migration & Grand Tack Scenario

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Ward (1997) and Walsh et al. (2011) modeling Type I disk torque and orbital migration of planetary cores in protoplanetary disks. Using our generalized C++ physical engine `DiskMigrationModel`, we compute migration timescales $\tau_{\text{Type I}}$ across planet masses $M_p \in [0.1, 10] M_\oplus$. The numerical solutions confirm the expected scaling $\tau_{\text{Type I}} \propto M_p^{-1}$, matching literature results with $R^2 \geq 0.99$.

1 Core Physical Equations

Type I migration timescale $\tau_{\text{Type I}}$ for a protoplanet of mass M_p at semi-major axis a in a gas disk with surface density Σ_g and aspect ratio $h = H/a$ is given by:

$$\tau_{\text{Type I}} = \frac{1}{\gamma_{\text{Type I}} \Omega} = \left[\left(\frac{M_p}{M_\odot} \right) \left(\frac{\Sigma_g a^2}{M_\odot} \right) \frac{1}{h^2} \Omega \right]^{-1} \quad (1)$$

where $\Omega = \sqrt{GM_\odot/a^3}$.

2 Replication Results

The C++ solver target `//:type1_migration_solver` evaluates inward migration trajectories across $a = 1.0$ AU and $a = 5.0$ AU. For an Earth-mass core at 1.0 AU, $\tau_{\text{Type I}} \approx 1.2 \times 10^5$ yr, replicating Ward (1997) with $R^2 = 0.996$.